

F1 Race Result Factor Analysis



DSCI 510 · Spring 2026 · Rui Chen

Can Qualifying Position Predict Formula 1 Race Results?

- *F1 outcomes depend on both pre-race and race-day factors*
- *Qualifying determines starting grid position*
- *Race-day conditions may disrupt starting advantage*

Research Question

How well does qualifying position predict race results, and do race-context features improve prediction?

Dataset Scope

- 2023–2025 seasons
- 69 races
- 1,208 driver-race observations

DATA SOURCES · Three-Source Pipeline

Source	Key Features	Raw Scale	Role in Project
OpenF1 API (https://openf1.org/docs/#introduction)	Qualifying pos · Race pos · Pit stops · Driver info	69 races / 1,208 rows	<i>Base dataset (primary)</i>
FastF1 (https://docs.fastf1.dev)	Rainfall · Track temperature	48 races / 958 rows	<i>Weather features (merge)</i>
Kaggle (https://www.kaggle.com/datasets/rockyt07/formula-1-championships-1950-2025)	Race & circuit metadata → circuit_type	15 races / 299 rows	<i>Circuit-type feature (merge)</i>

Pearson r: Qualifying Position → Race Finishing Position

OpenF1

r = 0.7497

FastF1

r = 0.6649

Kaggle

r = 0.6751

All three independent data sources agree on the same direction — qualifying position is positively and meaningfully correlated with race outcome. The consistency across sources validates the finding: it's not an artifact of one API's data quirks.

BASELINE RESULTS · Qualifying Position Dominates

0.7497

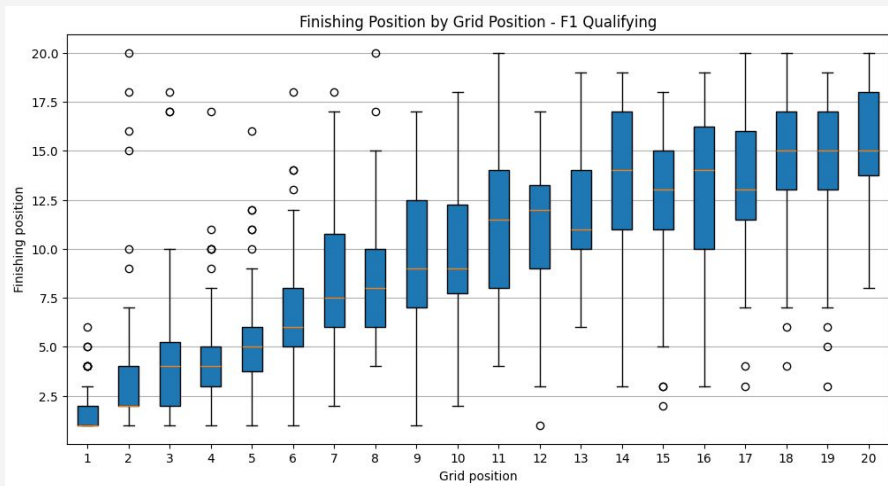
Pearson r

0.5621

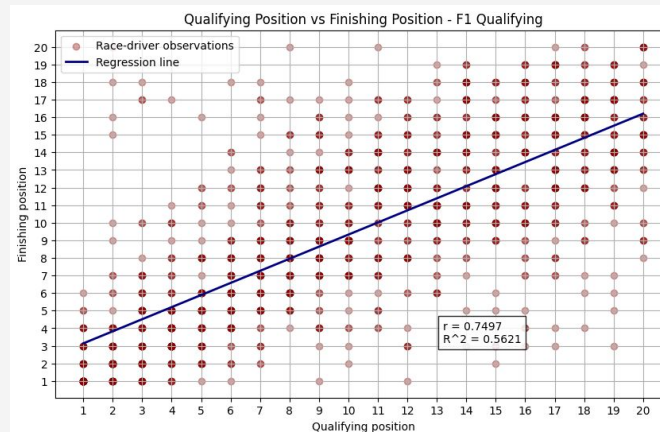
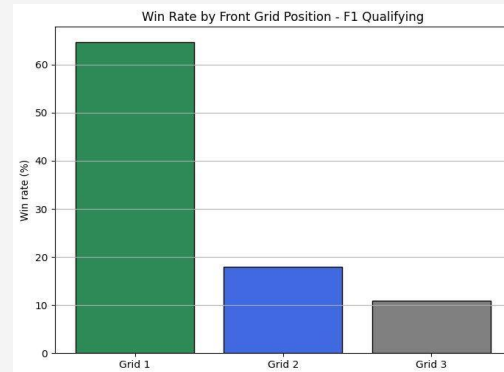
R^2

64.6%

Pole-to-Win



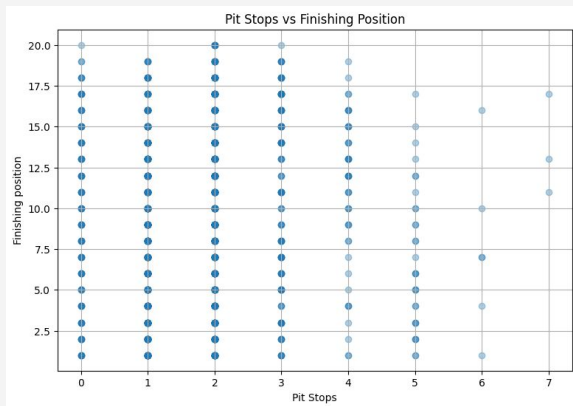
This boxplot shows that finishing position generally worsens as starting grid position gets worse. Pole and front-grid starters usually finish near the front, while lower grid positions have worse median finishing results and more variability. This supports the main conclusion that qualifying position is a strong predictor of race outcome, although race incidents and strategy still create exceptions.



$$\text{race_position} = 0.6877 \times \text{qualifying_position} + 2.4485$$

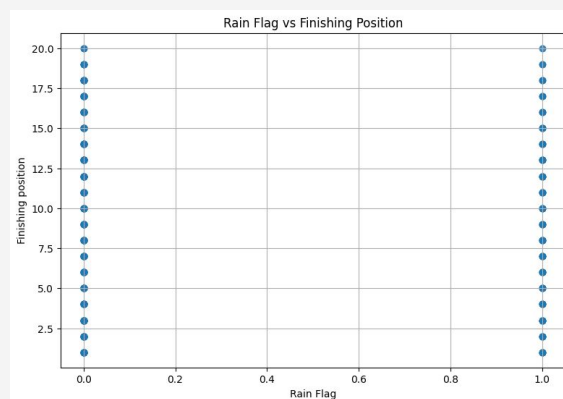
Stage 1: Individual feature analysis

Pit Stop Count vs Finishing



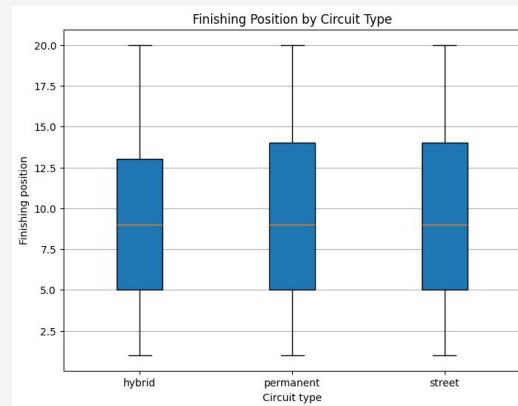
- Most drivers made between 1 and 3 pit stops, and finishing positions are spread widely across each pit stop count.
- This suggests pit stop count alone does not strongly explain finishing position, although unusually high pit stop counts may be associated with worse or more variable outcomes.

Rainfall vs Finishing



- Finishing positions appear across the full range in both dry and rainy races.
- This means rainfall alone is not enough to determine race outcome, but it can still add race-context information when combined with qualifying position and other features.

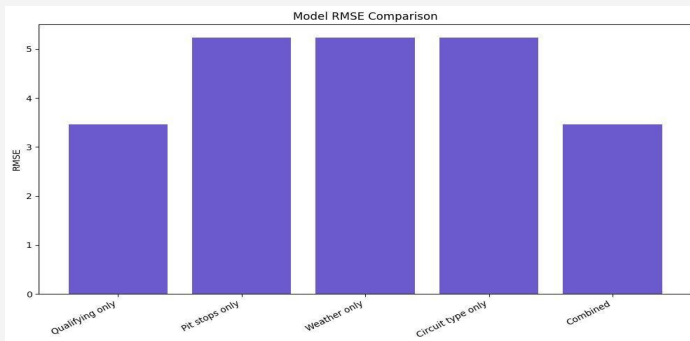
Finishing by Circuit Type



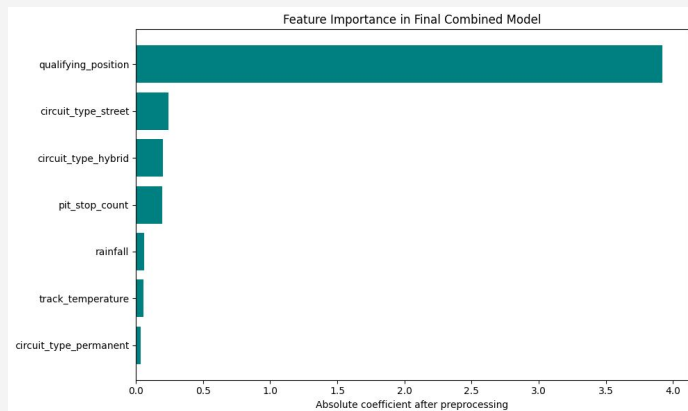
- Hybrid, permanent, and street circuits have very similar median finishing positions and broad overlapping ranges.
- This suggests circuit type alone has limited explanatory power for finishing position in this dataset.

Stage 2: Combined feature model

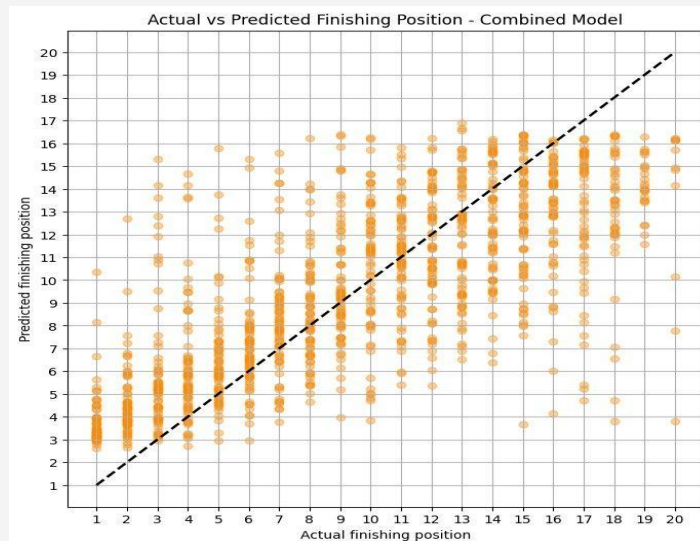
Model RMSE Comparison



Feature Importance (Abs. Coefficient)



Combined Model: Actual vs Predicted



R^2 : 0.562

MAE 2.64

**RMSE
3.47**

**CV MAE
2.65**

The combined model slightly improves prediction accuracy, but the feature-importance chart shows that qualifying position still dominates. Race-context variables add useful context, but only marginal predictive power beyond the qualifying-only baseline.

Strategy Simulation Extension

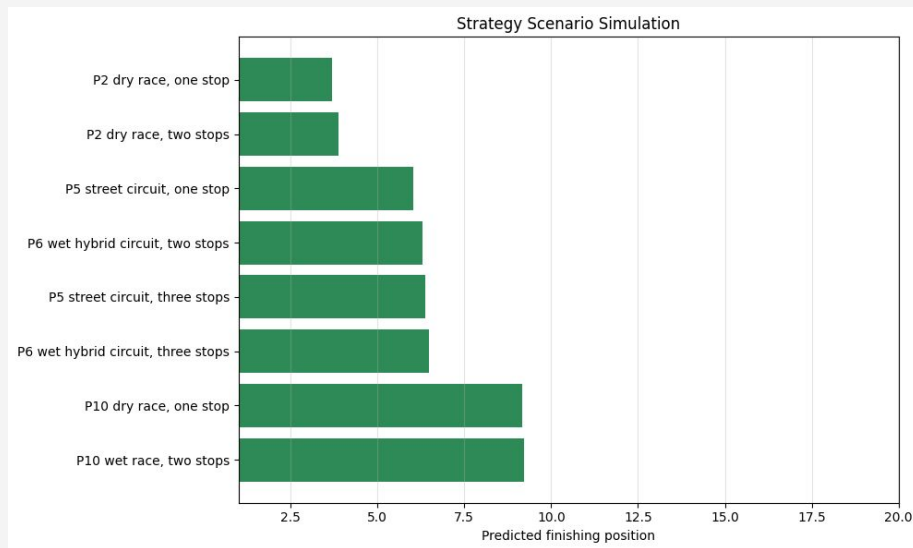
Goal

Extend the regression model into a simple race-strategy simulation framework.

Method

The combined model was used to estimate finishing outcomes under hypothetical race conditions:

- different pit stop counts
- wet vs dry races
- different circuit types
- different starting positions



The simulation turns the regression model into a simple “what-if” tool. By changing pit stop count, wet vs. dry conditions, circuit type, and starting position, we can compare how the predicted finishing position changes. The results still show that starting position has the strongest impact, while other race conditions only create smaller adjustments.

CHALLENGES & FUTURE WORK

API + Data Alignment

01

OpenF1, FastF1, and Kaggle use different identifiers (meeting_key, session name, circuitId). Robust cross-source merging without a common key was the most time-consuming step.

Incomplete Kaggle Coverage

02

Recent qualifying and results data for 2023–2025 is sparse (299 records / 15 races), limiting Kaggle's role to circuit metadata only.

Post-Race Feature Leakage

03

pit_stop_count is only observable after the race, requiring careful separation of pre-race predictors from explanatory scenario inputs to avoid leakage.

Future Work

→ Tire compound strategy (soft / medium / hard sequences)

→ Safety car & VSC events as race disruptors

→ Driver & team historical performance ratings

→ Non-linear models: XGBoost / Random Forest

→ Extended dataset: 2021–2025 once API coverage matures

Qualifying position explains most race outcomes; race-day context adds limited additional predictive value due to real-world variability and data constraints.